*Original Article*

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# The Deployment of Intersensory Selective Attention: A High-density Electrical Mapping Study of the Effects of Theanine

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## Abstract

**Objective:**
**Ingestion of the nonproteinic amino acid theanine (5-N-ethylglutamine) has been shown to increase oscillatory brain activity in the so-called alpha band (8–14 Hz) during resting electroencephalographic recordings in humans. Independently, alpha band activity has been shown to be a key component in selective attentional processes. Here, we set out to assess whether theanine would cause modulation of anticipatory alpha activity during selective attentional deployments to stimuli in different sensory modalities, a paradigm in which robust alpha attention effects have previously been established.**

**Methods:**
**Electrophysiological data from 168 scalp electrode channels were recorded while participants performed a standard intersensory attentional cuing task.**

**Results:**
**As in previous studies, significantly greater alpha band activity was measured over parieto-occipital scalp for attentional deployments to the auditory modality than to the visual modality. Theanine ingestion resulted in a substantial overall decrease in background alpha levels relative to placebo while subjects were actively performing this demanding attention task. Despite this decrease in background alpha activity, attention-related alpha effects were significantly greater for the theanine condition.**

**Conclusion:**
**This increase of attention-related anticipatory alpha over the right parieto-occipital scalp suggests that theanine may have a specific effect on the brain's attention circuitry. We conclude that theanine has clear psychoactive properties, and that it represents a potentially interesting, naturally occurring compound for further study, as it relates to the brain's attentional system.**

**Key Words: theanine, attention, attentional deployment, alpha oscillations, multisensory, EEG**

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Theanine (5-N-ethylglutamine) is a naturally occurring nonproteinic amino acid that, along with caffeine and cathecins, is 1 of 3 major constituents of tea (*Camellia sinensis*). Found only in tea and one rare mushroom variety (*Xerocomus badius*), it is relatively easy to extract from tea, and it is not uncommon for it to be consumed in its purified form as a dietary supplement, with many of the commercial concerns proffering claims regarding beneficial psychological and physiological effects. Regrettably, the vast majority of these claims remain to be properly substantiated in clinical trials. One claim that seems to have some reasonable measure of physiological support, however, relates to potential effects on the resting brain state. That is, ingestion of theanine has been shown to increase oscillatory brain activity in the so-called alpha band (8–14 Hz) when subjects are in a passive resting state.[1] Indeed, this brain rhythm has traditionally been associated with a relaxed state.[2] This finding of increased alpha activity because of theanine ingestion was recently corroborated by Nobre et al at the University of Oxford (personal communication May, 2003). An increase in alpha activity implies that theanine or one of its metabolites is likely psychoactive and warrants further investigation.

Activity in the alpha band has classically been associated with the general state

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of mental alertness or arousal,[3] implying in turn that theanine may exert effects on the level of alertness or attentiveness of humans. Indeed, anecdotal reports from tea drinkers often emphasize changes in general arousal and relaxation state. An open question, however, is whether the effects of theanine on oscillatory brain mechanisms are relatively nonspecific, limited to more general arousal states during periods of rest or passivity, or whether this compound also operates in a more specific manner on attentional control systems of the brain when subjects are engaged in more demanding and mentally taxing tasks.

Previous studies have shown that oscillations in the alpha band are not simply associated with brain arousal states, but that these oscillatory potentials are also key components in selective attentional processes.[4–15] That is, directed deployment of alpha oscillatory activity seems to underlie the suppression of distracting information in visual space during highly demanding attention tasks. For example, it has been shown that there is a highly specific pattern of alpha oscillatory activity when humans deploy their attention to stimulation in one sensory modality (eg, audition) while attempting to exclude or suppress inputs from an interfering sensory modality (eg, vision).[5,7,8]

In one such study, Foxe et al[5] used symbolic cue stimuli to instruct subjects to attend to a given sensory modality (either audition or vision). After a 1-second delay, a compound audiovisual multisensory stimulus appeared, and subjects performed a very demanding task solely within the cued modality while attempting to suppress inputs from the other distracting modality. It was shown that deploying attention to impending stimuli in the auditory modality resulted in a robust increase in alpha power during the late phase of the cue-target interval (CTI), that is, the attentional deployment phase, over parieto-occipital scalp regions. It was proposed that this oscillatory alpha enhancement reflected anticipatory gating or suppression of visual processing by parieto-occipital structures known to be involved in attentional switching and disengagement within the visual modality.[5,6,16,17] A subsequent study showed that similar alpha processes were deployed to locations in space where distracters might appear during a demanding visuospatial selective attention task,[6] reinforcing the role of alpha processes in attentional deployment and distracter suppression.[10,12]

The present study set out to assess whether theanine would cause modulation of anticipatory alpha activity during selective, attentional deployments to stimuli in different sensory modalities. Here, subjects performed a task similar to that of Foxe et al[18,19] in which they were instructed to deploy their attention to a given modality (auditory or visual) on a trial-by-trial basis. By inspecting the alpha band activity during the CTI, the effects of theanine on intersensory attentional deployments were assessed. A simple hypothesis was tested. We reasoned that because theanine is now known to affect overall alpha band activity during the resting state, it might also enhance (or indeed suppress) alpha-based attentional mechanisms. If the latter is the case, this would provide direct evidence that theanine does not simply have a general effect on arousal but can affect more specific attentional brain states. Furthermore, if theanine is found to enhance attentional processes, this compound might well represent a potentially attractive, naturally occurring, intervention for attentional disorders.

## METHODS

### Participants

Fifteen (eight women) neurologically healthy, paid volunteers (mean age, 27.8 years, SD ±6.5 years, range, 18–36 years) participated. All participants provided written informed consent, and the institutional review board of the Nathan Kline Research Institute approved the procedures. All participants reported normal or corrected-to-normal vision, and all were right-hand

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dominant as assessed by the Edinburgh Handedness Inventory.[20] Subjects were required to refrain from drinking any caffeine-based products (such as soft drinks, soda, coffee, or tea) for at least 24 hours before the day of testing. Subjects' neurological status was determined by conducting a shortened version of the Structured Clinical Interview for *Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition–Text Revision*.

## Experimental Paradigm

The sequence of events in a typical trial is illustrated in Figure 1. A trial commenced with the onset of the cue stimulus (S1) instructing participants as to which modality they should attend. Stimulus onset asynchrony between S1 and S2 (ie, the cue and the subsequent potential target) was 1200 milliseconds. The intertrial interval (ITI), that is, the period between each pair of cue-target stimuli, randomly varied between 1400 and 2400 milliseconds. A central fixation cross (black and 1-degree angle) remained on the screen throughout the experiment, and participants were instructed to maintain fixation at all times. All visual stimuli were displayed on a gray background, whereas all auditory stimuli were presented through headphones. Participants completed a minimum of 15 blocks of trials, on each of the 2 days of testing. Each block contained a total of 100 S1 to S2 pairs, giving an average block run time of 5

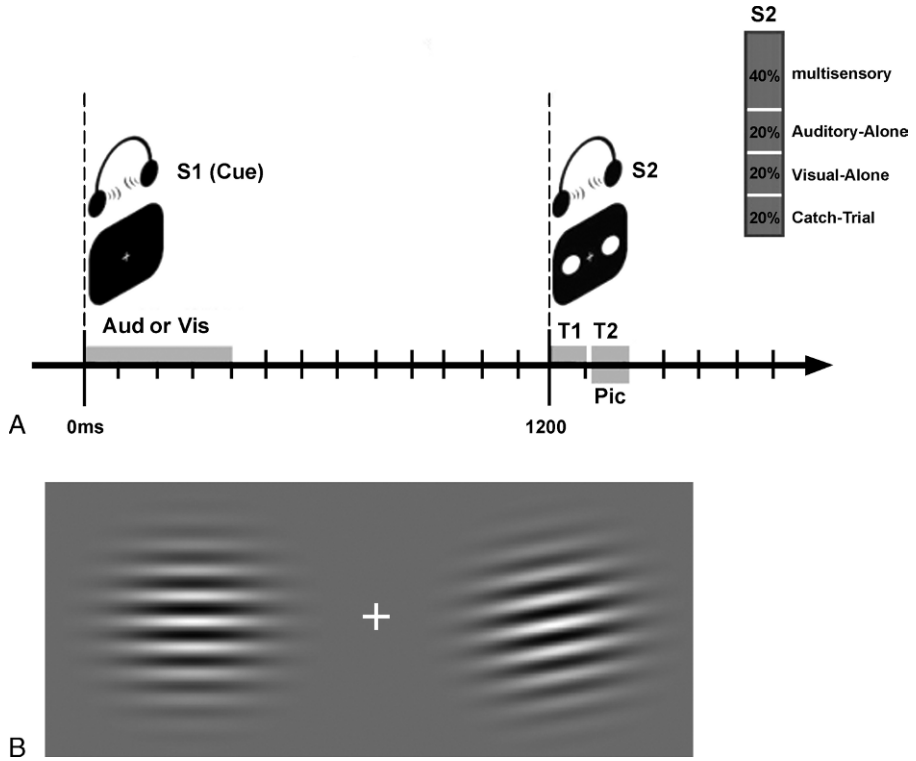


**FIGURE 1.** A, Schematic illustration of the paradigm. Each trial commenced with the presentation of an auditory symbolic cue (S1) whereby the Aud (for auditory) or Vis (for visual) were presented across headphones to the subjects. A delay period of 800 milliseconds followed the cue, after which 1 of 4 types of S2 stimuli would appear: auditory stimuli alone, visual stimuli alone, catch trials (where no physical stimulus was presented), or compound audiovisual multisensory stimuli. Subjects were required to respond with a push button to targets within the cued modality. B, An example of a visual target trial. A pair of Gabor patches was presented bilaterally with the orientation of one slightly different to the other. Target trials occurred on only 20% of visual S2s. On the other 80% of trials, the orientation of both patches was identical. The orientation difference was psychophysically calibrated on an individual basis such that subjects could only identify 80% of targets correctly.

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minutes. The specific details of the stimuli themselves follow.

**S1 (Cues)**

The cue stimulus (S1) was presented auditarily and was either the word "AUD" or "VIS," spoken by a male speaker and presented bilaterally via Sennheiser HD600 headphones ([Sennheiser Headphones, Old Lyme, Conn] duration, 400 milliseconds; 10-millisecond rise/fall, 80 dB sound pressure level). These word cues indicated the sensory modality (auditory or visual) to be attended when the second stimulus (S2) arrived. Specifically, the pseudoword "AUD" instructed participants to deploy their attention to the auditory modality, whereas the pseudoword "VIS" instructed the participants to attend to the visual modality. These cue-stimuli appeared in random order throughout the experiment.

**S2 (Targets)**

The S2 consisted of an auditory-alone (pair of tones, 20%), visual-alone (pair of Gabor patches, 20%), audio-visual compound (pair of tones and Gabor patches, 40%), or a null (catch) stimulus (no S2 stimulus, 20%). The auditory S2 stimulus consisted of a pair of binaural tones (eg, 2000 or 2100 Hz, 80 dB sound pressure level, 100-millisecond duration each, 5-millisecond rise/fall, 5-millisecond interval between tones). That is, the pair of tones was presented in a rapid sequence with auditory stimulation lasting for a total of 205 milliseconds. The visual stimulus consisted of a pair of Gabor patches (4.5 degrees in diameter, centered 2.5 degrees left and right of fixation, 100-millisecond duration). The audiovisual compound stimuli were a combination of the previously described auditory and visual stimuli. The null stimulus consisted of a period of fixation only. All visual stimuli were presented on an Iiyama VisionMaster Pro502 21-inch computer monitor on a gray background.

On 85% of visual S2 trials, the pair of Gabor patches was identical, and no overt response was required (ie, these were non-target standards). On 85% of auditory S2 trials, the pair of tones was also identical, and no response was required. On the other 15% of visual S2 trials, the orientations of the left and right Gabor patches were slightly different (see later), and subject responded with a push button (ie, these were target stimuli) when they had been cued to the visual modality. On 15% of auditory trials, the 2 tones were of slightly different pitch and also required a response when the auditory modality had been cued. The reader should note that the occurrence of a target in each modality was independent of the other modality, such that the probability of occurrence of a double target (ie, a bisensory target) was approximately 1%.

This scenario led to 4 possible target types: (1) unisensory-auditory (ie, when the cue instructed the subject to attend to the auditory modality, and only an auditory S2 stimulus occurred); (2) unisensory-visual, (3) multisensory-auditory (ie, when the cue instructed the subject to attend to the auditory modality, and an audiovisual compound multisensory S2 stimulus occurred), and (4) multisensory-visual.

## Procedure

At the beginning of each experimental day, all participants were given either a mixed solution of the theanine substance or a placebo drink. The mixed drink solution consisted of a mixture of 250 mg of powdered clear theanine with 200 mL of room temperature water. The placebo drink consisted only of the 200 mL of water (ie, approximately a cup of water). The day of drinking the theanine solution was counterbalanced across participants. Note that theanine is colorless and tasteless in a water solution and subjects were at chance in guessing whether they were taking the active compound or simply water.

On each experimental day, before the start of the testing phase, all participants performed a psychophysical test that equated the participant's performance to an 80% level on both the auditory and visual tasks to

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be used herein. This psychophysical test, known as the Up-Down Transformed Rule, is a rapid method that assures the experimenters that the level of task difficulty for each task will be equivalent across all participants.[21]

## Measurements

Continuous EEG data, digitized at 512 Hz, were acquired through the ActiveTwo Biosemi electrode system (Active Two Biosemi System: Amsterdam, Netherlands) from 168 scalp electrodes. BioSemi replaces the "ground" electrodes used in conventional EEG systems with 2 separate electrodes; the so-called Common Mode Sense active electrode and the Driven Right Leg passive electrode. These 2 electrodes form a feedback loop, which drives the average potential of the subject (the Common Mode voltage) as close as possible to the analog-to-digital converter (ADC) reference voltage in the AD box (the ADC reference can be considered as the amplifier "zero"). With the Biosemi system, every electrode or combination of electrodes can be assigned as the "reference," and this is done purely in software after acquisition. A detailed description of the referencing conventions used by this active electrode system can be found at the following Web site: http://www.biosemi.com/faq/cms&drl.htm.

All data were re-referenced to a midline prefrontal scalp site after acquisition. After each recording session, before the electrode cap was removed from the subject's head, the 3-dimensional coordinates of the electrodes with reference to anatomical landmarks on the head (nasion, preauricular notches) were digitized using a Polhemus magnetic 3-dimensional digitizer (Polhemus: La Jolla, Calif). This allowed the investigators to ensure that electrode locations were consistent between test days for all subjects. The EEG was recorded continuously and epoched and averaged off-line. Trials with blinks and large eye movements were rejected off-line on the basis of horizontal and vertical electrooculogram recordings. An artifact rejection criterion of ±100 μV was used at all other electrode sites to exclude periods of high EMG and other noise transients.

## Data Analysis Strategy

Accepted trials were epoched for the period around the onset of the S1 cues (−300 milliseconds prestimulus to 1200 milliseconds poststimulus). Note that only the S1 stimuli are analyzed for the present study. The baseline was defined as the mean voltage from 300 to 0 millisecond before the onset of S1. Separate averages were made for the 2 possible variants of the S1 stimuli (cue-AUD and cue-VIS). We inspected oscillatory activity in the alpha band (8–14 Hz[1]) during the CTI. Alpha band activity was characterized in this period by the temporal spectral evolution (TSE) technique, which provides an index of induced alpha activity as a function of time (see Fu et al[7] for a full description of the method). All statistical analyses were performed on these induced alpha oscillations. The TSE waveforms are derived by the following method:

- Individual (single trial) stimulus-locked epochs are band-pass filtered after artifact rejection (Butterworth zero-phase, 8–14 Hz, 48 dB/octave).
- Filtered epochs are then full-wave rectified (ie, all negative data points are made positive).
- Rectified waveforms are then averaged.

Repeated-measures analysis of variance (ANOVA) was used to statistically test for effects over a region of interest (ROI) defined by 4 electrode sites over the right parieto-occipital scalp (the ROI chosen here was based on previous work from this laboratory, where maximal alpha-dependent attention effects have been defined[5]). Factors were attention condition (auditory vs visual) and treatment (theanine vs placebo). The

[1] The exact band-pass that constitutes the alpha band is not consistent across the literature and could be considered somewhat arbitrary. In fact, the centre frequency of alpha is quite variable across individuals and for most it tends to be in the 10–12 Hz range.[22] As such, the band-pass chosen here of 8–14 Hz nicely spans this range.

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epoch chosen for testing here (700– 1100 milliseconds) was also based on our previous work,[5–7] where it was shown that the alpha effect was maximal in the latter part of the CTI in the period preceding the onset of the S2 (ie, for approximately 400 milliseconds). The dependent measure was an integrated area measure across this 400-millisecond epoch—that is, how far was the TSE waveform displaced from zero.

A secondary analysis of the relative attentional differences between treatment conditions in induced oscillations was also performed. Specifically, we derived an attentional modulation index. This technique is a useful metric to normalize the data set with regard to background alpha levels. Modulation indices on the induced alpha oscillations were derived by the following method:

- Step 1: An average background alpha activity value was computed for each treatment condition across the 1200-millisecond CTI.
- Step 2: After computing this average, the difference in alpha amplitude between the attend-auditory and attend-visual condition during the 400-millisecond period from 700 to 1100 milliseconds was divided by the appropriate mean value computed in step 1.

## RESULTS

### Performance Data

Percent correct-hit responses (accuracy) and reaction time (RT) mean values are shown in Figure 2. Two 2 × 4 repeated-measures ANOVAs with factors of treatment (placebo vs theanine) and S2 target type

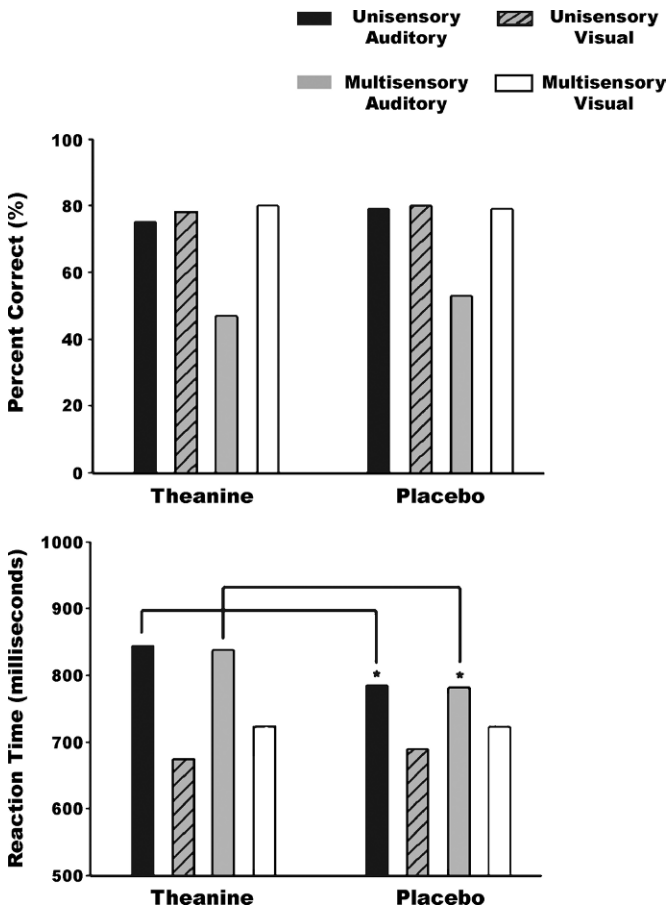


**FIGURE 2.** Behavioral data. Hit rates are plotted for each treatment in the upper graph. Mean reaction times for correct responses are plotted in the lower graph. Asterisks illustrate statistically significant effects between conditions.

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(unisensory-auditory, multisensory-auditory, unisensory-visual, multisensory-visual) were computed for both the accuracy and RT data.

For accuracy, a main effect of stimulus type was found ($F_{3,45}$ = 12.364, $P < 0.01$), driven by a considerable performance decrement that was seen for the multisensory-auditory stimulus type during both treatment conditions. The reader will recall that the auditory task is identical between the unisensory-auditory and multisensory-auditory conditions, so the presence of distracting visual information in the latter case had a large impact on performance accuracy. No other effects reached significance—that is, accuracy was not found to be different in any condition between treatments (ie, placebo vs theanine).

For the RT data, a main effect of stimulus type was found ($F_{1,14}$ = 12.23, $P < 0.01$) caused mainly by faster RTs to the unisensory-visual stimuli, again observed during both treatment types. An interaction of treatment × stimulus type ($F_{3,45}$ = 5.96, $P < 0.05$) was also seen. This was because RTs were considerably slower during the theanine treatment than placebo for both the unisensory-auditory and multisensory-auditory stimulus types. No other significant effects were found.

## Induced Alpha Band Activity

Alpha band activity during the attentional deployment phase is illustrated in Figure 3. Depicted are average 8- to 14-Hz TSE waveforms across the four electrode sites chosen for testing over the right parieto-occipital scalp region. It can be seen that the divergence in the alpha band activity starts to occur at approximately 400 milliseconds and is sustained until the onset of the S2 stimulus at 1200 milliseconds, where the largest difference is observed.

A repeated-measures ANOVA performed on these TSE waveforms revealed a

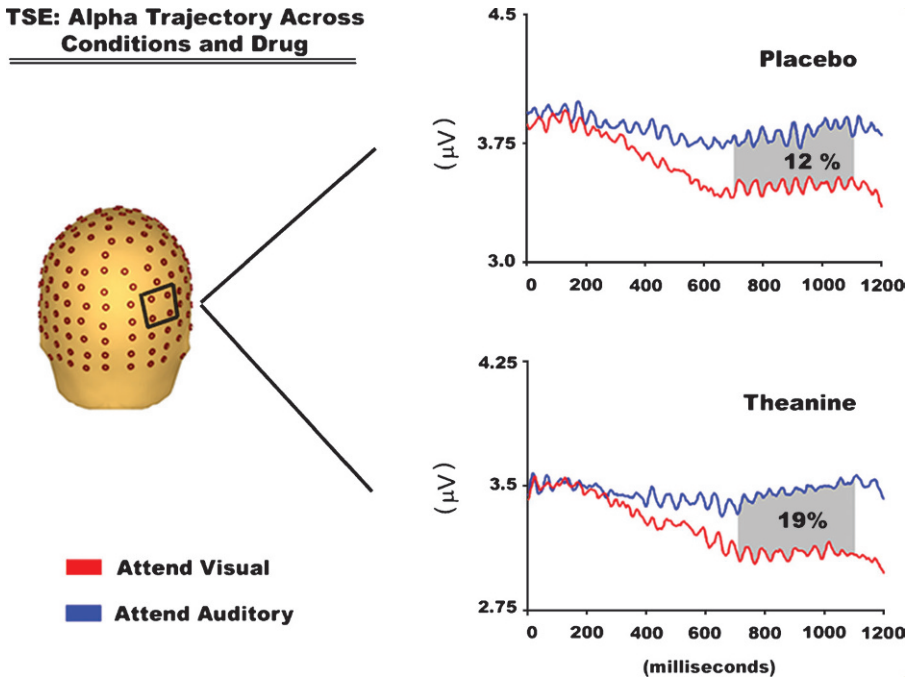


**FIGURE 3.** Alpha band oscillatory activity is selectively modulated by the deployment of anticipatory attention to different sensory modalities. The TSE waveforms, averaged from the 4 scalp sites illustrated over the right parieto-occipital scalp, are plotted for placebo (upper panel) and theanine conditions (lower panel). A sustained divergence in TSE amplitude is seen starting at approximately 400 milliseconds after cue. In both treatment conditions, the alpha band activity is significantly greater when subjects have been cued to attend selectively to impending auditory stimulation (blue trace) compared with visual stimulation (red trace). However, this sustained divergence is significantly greater during the theanine condition compared with the placebo condition.

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main effect of treatment ($F_{1,14}$ = 9.112, $P$ < 0.01), indicating that the overall alpha activity in the late sustained period is significantly reduced by the theanine solution. Figure 3 shows that this drop in the alpha power is present across the whole CTI, not just in the late sustained period.

The ANOVA also revealed a main effect of attention ($F_{1,14}$ = 9.776, $P$ < 0.01). A trend toward an interaction of treatment × attention was also observed ($F_{1,14}$ = 3.056, $P$ = 0.1). No other effects reached significance.

In the second phase of our analysis, we derived an attention modulation index. After the computation of these indices, a paired $t$ test between the treatment conditions (collapsed across attention conditions) was computed on the same ROI mentioned previously. The $t$ test revealed a significant difference ($t_{14}$ = −2.465, $P$ = 0.027), indicating that the overall amplitude of the alpha attention effect as a function of background alpha was significantly greater for the theanine treatment day than the placebo day.

To investigate the topographic distribution of the induced alpha effect, the TSE response of the attend-visual condition was subtracted from the TSE response to the attend-auditory condition. This subtraction results in a derived response representing only the differential activity between attention conditions. After this subtraction, the resulting waveform was modeled using the minimum norm solution technique of the BESA 5.01 software (Brain Electric Source Analysis, Gräfelfing, Germany. Available at: www.besa.de).

The topographic analysis of the induced alpha band oscillatory effect reveals a distribution over bilateral parieto-occipital regions, with a predominance of activity over the right hemisphere (Fig. 4). The theanine treatment condition elicits a greater differential and sustained alpha effect over the period.

## DISCUSSION

The present study showed that theanine consumption caused substantial effects on the brain's generation of the alpha oscillatory rhythm (8–14 Hz) while subjects performed a demanding intersensory selective attention task. First, theanine consumption resulted in a substantial decrease in overall background alpha power, regardless of which sensory modality was being attended. Second, theanine caused a relative amplitude increase of a previously defined alpha attention effect associated with the deployment of intersensory selective attention.[5,7] These earlier studies showed that cuing attention to the auditory features of an imminent, compound audiovisual stimulus resulted in significantly higher alpha amplitude in the period preceding the onset of this stimulus than when attention was cued to the visual features. Here, we found that this difference in anticipatory alpha activity was relatively greater when subjects had ingested theanine, suggesting that theanine may have enhanced this well-characterized attention effect. This anticipatory alpha activity is generated within the right parieto-occipital cortex, a critical node of the brain's attention circuitry, further suggesting that theanine, rather than having a more general effect on arousal levels, may be having more specific effects on the brain's attention circuitry. These and other implications of the present data are discussed in what follows.

### A Resting Versus Active Alpha Paradox

The motivation for the present study was that theanine has been shown to increase ongoing tonic alpha power during the resting or passive state. That is, the 2 previous studies of the neurophysiology of theanine in humans have simply recorded ongoing EEG while subjects sat passively and were not performing any cognitive task.[1] That theanine might enhance alpha activity is of real interest, as increased alpha amplitude may have implications for general cognitive performance abilities. For example, it has been shown that children with

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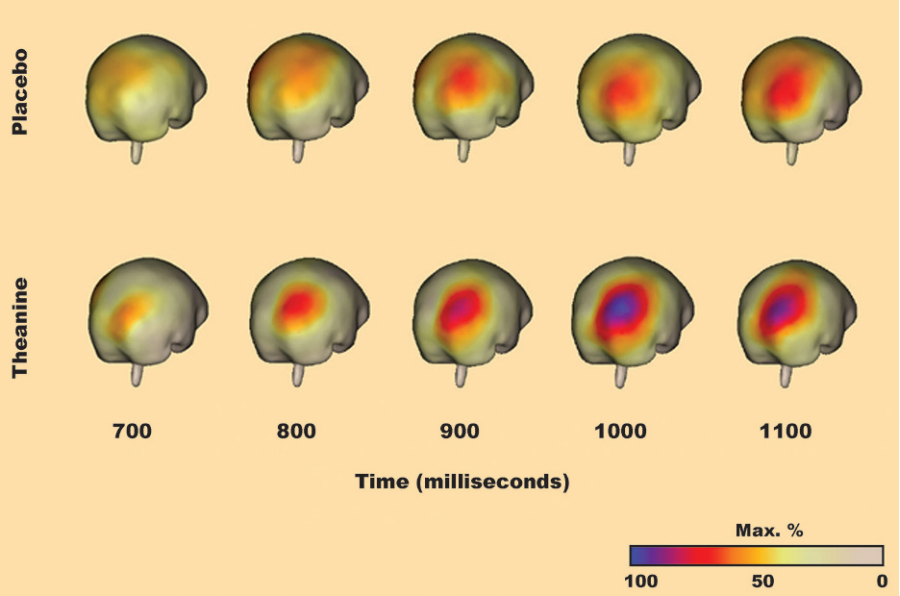


**FIGURE 4.** Topographic analysis using the minimum norm solution reveals a clear predominance of alpha oscillatory activity over the right parieto-occipital region. The theanine treatment condition (lower field) elicits a greater differential and sustained alpha effect compared with placebo (upper field).

an impoverished educational background and low reading and writing scores show generally reduced resting alpha power.[23] The same appears to be true of children with attention deficit/hyperactive disorder who show generally reduced alpha power,[24] and evidence also suggests that this reduction can be normalized through medication.[25,26] A relationship between increasing alpha across practice sessions and improved performance on a verbal working memory task has also been shown.[27] Most recently, it has been shown that performance on a visuospatial attention task was predicated on the amplitude of preceding anticipatory alpha.[13] In a design closely modeled on the original one used by Worden et al,[6] subjects were cued by a central arrow to attend to either the left or right hemifields for an impending visual target. Reaction times to detected targets were faster on the attended side, and for both left and right targets, RTs varied as a function of the amplitude of anticipatory alpha, with faster RTs to right targets when alpha was greater over the right hemisphere (lower on the left) and faster RTs to left targets when alpha was greater on the left (lower on the right). Overall, there seems to be growing evidence that increased tonic alpha power (ie, during the resting state) may be related to better cognitive performance abilities, and that the strength of alpha processes during more active tasks may be a predictor of performance on demanding attentional tasks.

Although the present study was designed to specifically assess alpha attention effects, 1 surprising and apparently paradoxical outcome was that overall background power was significantly lower in the theanine treatment condition compared with placebo. It is important to point out that this reduction in alpha power is only seen while subjects are engaged in a very demanding cognitive task. In this sense, our study is entirely different from the previous EEG studies. Unfortunately, in the present study, a measure of tonic alpha was not taken during a resting condition as in the 2 previous studies, which is why a direct comparison between these results and previous work is not possible. It should also be noted

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that despite the overall drop in background alpha observed here, this had no effect on the subjects' ability to effectively deploy alpha attention mechanisms because the differential alpha effect was greater in the theanine condition. That is, although background alpha dropped in amplitude, concurrent attention-related alpha processes were enhanced. As such, these results may suggest that overall background alpha and attention-related alpha processes represent distinct cognitive functions, a possibility meriting further investigation.

Another consideration here is the makeup of the subject group in our study relative to the previous studies, which were carried out in Japan[1] and England (Nobre Laboratory). The young American adults who served in our study were not regular tea drinkers, and as such, had little to no previous exposure to theanine. This is unlikely to have been the case in the previous studies as tea drinking is considerably more prevalent in both countries than it is in the United States. Levels of previous and ongoing exposure to theanine are factors that will have to be accounted for in future studies of this compound.

## Visual Dominance During Multisensory Target Trials

One dramatic effect seen here in behavioral performance (for both the placebo and theanine conditions) was a substantial reduction in performance accuracy for the auditory task when visual stimuli were concurrently presented—that is, during the audiovisual multisensory target condition. The reader will recall that both the auditory and visual tasks were psychophysically equated before any recordings were made, with performance calibrated such that all subjects detected exactly 80% of targets in both sensory modalities. However, this pretest psychophysical calibration was only conducted for the unisensory stimulus conditions. Behavioral results show that the manipulation was fully effective as performance remained at or very close to 80% for both unisensory stimulus conditions across both treatment days. The same result extends to the multisensory-visual targets, where subjects also achieved about 80% accuracy, indicating that the presence of simultaneous distracting auditory inputs had no effect on performance of the visual task. The reverse was not the case. Subjects' performance dropped substantially for the auditory task when concurrent distracting visual inputs were present (from 80% to approximately 50%).

This finding has implications for 1 prominent view in attentional theory. A central tenet of modern theory is that selective attention is necessary because of the limited informational processing capacity of the brain.[28] That is, selective attention functions to filter the overwhelming quantity of information constantly impinging on our senses, allowing for preferential processing of a subset of these inputs. Implicit in this construct is the notion that stimuli that are not in the attentional focus receive reduced processing. The existence of a limited capacity is seen in divided attention tasks, where subjects monitor 2 streams simultaneously, and clear deficits in performance within a given stream are seen because of interference from the second stream.[29,30] However, some have claimed that this capacity limitation only occurs when the 2 streams of information are within a single sensory modality, and that no such capacity limit is present when the streams of information occur within separate sensory systems. For example, in a seminal article, Duncan et al[31] used the so-called attentional blink (AB) paradigm to show that there was no interference between modalities. In the AB paradigm, subjects are typically required to respond to 2 target stimuli that occur in relatively close temporal proximity within a stream of distracters. Identification of the first target in the sequence produces a sustained reduction in the ability to identify the second target stimulus when the 2 targets are presented within a single modality—that is, when all

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stimuli are only visual (or only auditory). In contrast, Duncan et al[31] showed that when a pair of targets was presented in 2 different modalities, one auditory and the other visual (or vice-versa), no such interference effects were found. Similar results for audiovisual pairings have been found[2].[32,33] This finding led Duncan et al[31] to propose that "visual attention to one simple target does not restrict concurrent auditory attention to another," and to conclude that there was no restricted attentional capacity between sensory modalities.

The present results argue otherwise. Concurrent visual stimulation, despite being completely irrelevant to the task at hand, caused a substantial decrement in subjects' abilities to perform the auditory discrimination task, suggesting that supramodal (suprasensory) attentional systems do in fact have a limited capacity. In this respect, our results are reminiscent of early work by Colavita,[35] Colavita and Weisberg,[36] and Sinnett et al.[37]

The question becomes why such a strong interference effect is seen in these data but not in the AB paradigms of Duncan et al.[31] One possible explanation is that a degree of automatic multisensory integration of the simultaneous auditory and visual constituents of our multisensory targets occurred. Indeed, Duncan et al[31] explicitly pointed out that their auditory and visual stimulation streams were clearly separable, and they were careful to specify that other circumstances might apply for the case where the auditory and visual events emanated from the same object (ie, when multisensory integration might occur). Our auditory and visual stimuli did occur in temporal register, although it should be emphasized that they were presented in different spatial locations (headphones vs computer screen), and that they had no natural relationship with each other. Furthermore, subjects were required to explicitly treat them as separate stimulus streams to be attended differentially, and this separability would have been reinforced by the fact that only 40% of the S2 stimuli were actually bisensory. Nonetheless, some degree of automatic multisensory integration of the auditory and visual components of S2 may well have occurred, and there is precedence in the literature for integration of even such basic stimuli.[38–41]

Another obvious question that arises is why the interference effect is found to be unidirectional, only affecting performance on the auditory task when targets were bisensory, with visual performance remaining unaffected. A striking phenomenon in the study of perceptual models of multisensory stimuli is the tendency of visual information to dominate other competing sensory stimuli. Jordan[42] and Posner et al[43] have argued that visual dominance phenomena arise from brain mechanisms designed to compensate for the "low alerting capability of visual signals." The premise is that because of this intrinsically lower state of alertness in the visual system, the attentional system is biased toward vision when there is a probability that reliable information will be provided in this modality. As a consequence, this visual biasing would result in the withdrawal of processing resources from other senses. It should be pointed out though that Posner et al[43] proposed that this general attentional biasing would only occur when the subjects are likely to receive reliable inputs from the visual modality. To test this, Klein (unpublished data, 1974) presented subjects with random trials of visual, somatosensory, or multisensory somatovisual stimuli. He found that subjects responded fastest to the multisensory stimuli and slowest to the visual-alone stimuli. These results led Posner et al[43] to propose that in a conflict situation (ie, when vision, proprioception, and/or audition provide discrepant information), visual inputs will tend to dominate the brain's attentional system, unless the subject is aware that responses based on visual

[2] Note that an AB was in fact found by Soto-Faraco et al[34] between the visual and somatosensory modalities, suggesting that capacity limitations may vary across sensory combinations.

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information will be disadvantageous.[43] Both Jordan[42] and Posner et al[43] predicted that visual dominance would no longer prevail under circumstances where vision does not provide adequate information.

The present results simply do not align with this view. A strong visual dominance effect persists, even when the visual stimuli are known by subjects to be immaterial to the task at hand, suggesting that endogenous attentional processes cannot fully supersede exogenous attentional capture by visual inputs. Our data suggest that there is indeed an attentional capacity limit for streams of information occurring across 2 sensory modalities. The contention that there is no attentional capacity limit between sensory modalities is also not supported by data from patients with spatial neglect syndrome. For example, cross-sensory extinction between vision and touch in neglect patient has been shown,[44] again suggesting a central supramodal attentional bottleneck.[45,46] Similarly, extinction between audition and touch has been described.[47]

### Theanine Slows RTs to Auditory Stimuli

Our results also showed a marked slowing of RTs for all auditory conditions when subjects had ingested theanine (by as much as 60 milliseconds), although there was no change in performance accuracy. This fairly dramatic slowing of responses was unique to the auditory modality, as RTs to all visual stimuli remained unaffected by theanine. One possibility is that theanine may cause subjects to be more deliberate when confronted with such stimuli, although why this would only occur for the auditory stimuli is not clear, and if subjects have become more deliberate, they certainly have not benefited in terms of accuracy from such a change. Whatever the case, and this needs more study, theanine appears to differentially affect the auditory and visual sensory modalities, causing a measurable slowing of performance in the auditory domain. One possibility is that theanine may increase susceptibility to visual dominance as described previously.

## CONCLUSIONS

High-density electrical mapping showed that theanine ingestion had robust effects on alpha oscillatory activity in the cortex of healthy adult subjects while they performed a highly demanding intersensory attention task. We found a robust decrease in background alpha activity but a concomitant increase in attention-related alpha effects; that is, during the late sustained period of an attentional deployment phase, while subjects prepared to attend to either the visual or auditory modality for an impending multisensory target, differential alpha deployments were observed with greater amplitude than was seen during a placebo control condition. Topographic mapping confirmed that these alpha processes were generated in the posterior parietal cortices, especially over the right hemisphere. Thus, theanine, as well as affecting more general brain rhythms associated with cortical arousal, was found to have more specific effects on known nodes of the visual attention circuit. We conclude that theanine has clear psychoactive properties, and that it represents a potentially interesting, naturally occurring compound for further study as it relates to the brain's attentional system.

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